

Dynamics of Open Quantum Systems II: Reduced Density Operator and Quantum Master Equation

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In the previous note we studied the density operator for quantum systems. Applying its equation of motion to the transition between two manifold of states leads to the master equation and the golden rule formula. Such derivation relies on a specific model Hamiltonian:

$$\hat{H} = \sum_a E_a |a\rangle\langle a| + \sum_\alpha E_\alpha |\alpha\rangle\langle\alpha| + \sum_a \sum_\alpha (V_{a\alpha} |a\rangle\langle\alpha| + V_{\alpha a} |\alpha\rangle\langle a|)$$

where we assumed that **the interstate coupling is weaker than the system-reservoir coupling, so that the relaxation of reservoir is faster than the dynamics of the system.** Furthermore, to arrive at the (generalized) master equation, we have ignored the off-diagonal elements of the density matrix in certain steps of the derivation, therefore the final expression contains **no coherence between the initial and final states.**

Due to these approximations, the accurate **dissipation and dephasing dynamics can not be properly described by the golden rule formula.** To overcome this limitation, in this note, we will derive the quantum master equation (QME) based on the reduced density operator (RDO) for the system coupled with a reservoir.

1 Reduced Density Operator

The total Hamiltonian $\hat{H}$ of a system coupled with a reservoir can be separate to a system part $\hat{H}_S$, a reservoir part $\hat{H}_R$, and an interaction part $\hat{H}_{SR}$,

$$\hat{H} = \hat{H}_S(\mathbf{s}) + \hat{H}_R(\mathbf{Z}) + \hat{H}_{SR}(\mathbf{s}, \mathbf{Z}) \quad (1)$$

where $\mathbf{s}$ and $\mathbf{Z}$ are the system and reservoir degrees of freedom (DOFs), respectively. The density operator of the total system is

$$\hat{W}(t) = \sum_\mu w_\mu |\Psi_\mu(t)\rangle\langle\Psi_\mu(t)| \quad (2)$$

and the density matrix in coordinate representation is

$$\rho(\mathbf{s}, \mathbf{Z}, \mathbf{s}', \mathbf{Z}', t) = \sum_\mu w_\mu \Psi_\mu(\mathbf{s}, \mathbf{Z}, t) \Psi_\mu^*(\mathbf{s}', \mathbf{Z}', t) = \langle \mathbf{s}, \mathbf{Z} | \hat{W}(t) | \mathbf{s}', \mathbf{Z}' \rangle \quad (3)$$

Generally we only care about the dynamics of the system and don't want an explicit description of the reservoir. Let $\hat{O} = \hat{O}(\mathbf{s})$ be an operator that only act on the system S. The expectation value of $\hat{O}$ is

$$\begin{aligned}
\langle \hat{O}(\mathbf{s}) \rangle &= \text{Tr} \left(\hat{W} \hat{O}(\mathbf{s}) \right) \\
&= \int d\mathbf{s} d\mathbf{Z} \int d\mathbf{s}' d\mathbf{Z}' \langle \mathbf{s}, \mathbf{Z} | \hat{W}(t) | \mathbf{s}', \mathbf{Z}' \rangle \langle \mathbf{s}', \mathbf{Z}' | \hat{O}(\mathbf{s}) | \mathbf{s}, \mathbf{Z} \rangle \\
&= \int d\mathbf{s} d\mathbf{Z} \int d\mathbf{s}' d\mathbf{Z}' \langle \mathbf{s}, \mathbf{Z} | \hat{W}(t) | \mathbf{s}', \mathbf{Z}' \rangle \langle \mathbf{s}' | \hat{O}(\mathbf{s}) | \mathbf{s} \rangle \langle \mathbf{Z}' | \mathbf{Z} \rangle \\
&= \int d\mathbf{s} d\mathbf{Z} \int d\mathbf{s}' d\mathbf{Z}' \langle \mathbf{s}, \mathbf{Z} | \hat{W}(t) | \mathbf{s}', \mathbf{Z}' \rangle \langle \mathbf{s}' | \hat{O}(\mathbf{s}) | \mathbf{s} \rangle \delta(\mathbf{Z} - \mathbf{Z}') \\
&= \int d\mathbf{s} d\mathbf{s}' \left(\int d\mathbf{Z} \langle \mathbf{s}, \mathbf{Z} | \hat{W}(t) | \mathbf{s}', \mathbf{Z} \rangle \right) \langle \mathbf{s}' | \hat{O}(\mathbf{s}) | \mathbf{s} \rangle
\end{aligned} \tag{4}$$

where we have used the relation $\langle \mathbf{Z}' | \mathbf{Z} \rangle = \delta(\mathbf{Z} - \mathbf{Z}')$. The integral in the parenthesis in the last line is an averaged density matrix over the reservoir DOF $\mathbf{Z}$, known as the **reduced density matrix**,

$$\rho(\mathbf{s}, \mathbf{s}', t) = \int d\mathbf{Z} \rho(\mathbf{s}, \mathbf{Z}, \mathbf{s}', \mathbf{Z}, t) = \langle \mathbf{s} | \int d\mathbf{Z} \langle \mathbf{Z} | \hat{W}(t) | \mathbf{Z} \rangle | \mathbf{s}' \rangle = \langle \mathbf{s} | \hat{\rho}(t) | \mathbf{s}' \rangle \tag{5}$$

The **reduced density operator** (RDO) for the system is defined as

$$\boxed{\hat{\rho}(t) = \int d\mathbf{Z} \langle \mathbf{Z} | \hat{W}(t) | \mathbf{Z} \rangle = \text{Tr}_R \hat{W}(t)} \tag{6}$$

where Tr_R is the partial trace that only perform for the reservoir. Clearly,

$$\langle \hat{O}(\mathbf{s}) \rangle = \text{Tr}_S \left(\hat{\rho} \hat{O}(\mathbf{s}) \right) \tag{7}$$

The equation of motion for the RDO can be obtained by taking the partial trace Tr_R on both sides of the quantum Liouville equation,

$$\begin{aligned}
\frac{\partial}{\partial t} \hat{\rho}(t) &= -\frac{i}{\hbar} \text{Tr}_R \{ [\hat{H}, \hat{W}(t)] \} \\
&= -\frac{i}{\hbar} \text{Tr}_R \{ [\hat{H}_S, \hat{W}(t)] \} - \frac{i}{\hbar} \text{Tr}_R \{ [\hat{H}_R, \hat{W}(t)] \} - \frac{i}{\hbar} \text{Tr}_R \{ [\hat{H}_{SR}, \hat{W}(t)] \}
\end{aligned} \tag{8}$$

Since $\hat{H}_S$ does not act on the reservoir, it can be taken out of the partial trace, therefore the first term becomes

$$\text{Tr}_R \{ [\hat{H}_S, \hat{W}(t)] \} = [\hat{H}_S, \text{Tr}_R \hat{W}(t)] = [\hat{H}_S, \hat{\rho}(t)] \tag{9}$$

And the second term is zero because

$$\text{Tr}_R (\hat{H}_R \hat{W}(t)) = \text{Tr}_R (\hat{W}(t) \hat{H}_R) \tag{10}$$

The equation of motion for RDO thus becomes

$$\boxed{\frac{\partial}{\partial t} \hat{\rho}(t) = -\frac{i}{\hbar} [\hat{H}_S, \hat{\rho}(t)] - \frac{i}{\hbar} \text{Tr}_R \{ [\hat{H}_{SR}, \hat{W}(t)] \}} \tag{11}$$

When there is no interaction between system and reservoir, $\hat{H}_{\text{SR}} = 0$, Eq. (11) reduces to the quantum Liouville equation for the system only. When $\hat{H}_{\text{SR}} \neq 0$, Eq. (11) is not a closed equation for $\hat{\rho}$ because the right-hand side contains the full density operator $\hat{W}$. A closed equation can be obtained by taking perturbative expansion with respect to $\hat{H}_{\text{SR}}$. This perturbation approximation can be done more conveniently using the interaction picture of quantum mechanics.

2 Interaction Picture

In Schrödinger picture of quantum mechanics, the operators are time-independent, while the quantum states evolve with time. The solution of the time-dependent Schrödinger equation can be expressed in terms of the time evolution operator $\hat{U}(t)$,

$$|\Psi^{(\text{S})}(t)\rangle = \hat{U}(t)|\Psi^{(\text{S})}(0)\rangle \quad (12)$$

The expectation value of the operator $\hat{A}$ is

$$\langle \hat{A} \rangle = \langle \Psi^{(\text{S})}(t) | \hat{A}^{(\text{S})} | \Psi^{(\text{S})}(t) \rangle = \langle \Psi^{(\text{S})}(0) | \hat{U}^\dagger(t) \hat{A}^{(\text{S})} \hat{U}(t) | \Psi^{(\text{S})}(0) \rangle \quad (13)$$

In Heisenberg picture, the operators evolves with time while the quantum states are time-independent. If we define

$$\hat{A}^{(\text{H})}(t) = \hat{U}^\dagger(t) \hat{A}^{(\text{S})} \hat{U}(t) \quad (14)$$

and

$$|\Psi^{(\text{H})}\rangle = |\Psi^{(\text{S})}(0)\rangle \quad (15)$$

Clearly the expectation value of the operator $\hat{A}$

$$\langle \hat{A} \rangle = \langle \Psi^{(\text{H})} | \hat{A}^{(\text{H})}(t) | \Psi^{(\text{H})} \rangle \quad (16)$$

is the same in both pictures. Using the equation of motion of the time evolution operator,

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t) = \hat{H} \hat{U}(t) \quad (17)$$

We have the equation of motion for the operator in Heisenberg picture,

$$\frac{d}{dt} \hat{A}^{(\text{H})}(t) = \frac{i}{\hbar} [\hat{H}, \hat{A}^{(\text{H})}(t)] \quad (18)$$

Note that this equation assumes that $\hat{A}^{(\text{S})}$ is time-independent.

In the interaction picture, suppose we can separate the total Hamiltonian $\hat{H}$ to a stationary part $\hat{H}_0$ and a perturbation $\hat{V}$. $\hat{V}$ can be time-dependent. The quantum state in the interaction picture is defined as

$$|\Psi^{(\text{I})}(t)\rangle = \hat{U}_0^\dagger(t) |\Psi^{(\text{S})}(t)\rangle \quad (19)$$

where $\hat{U}_0 = \exp(-\frac{i}{\hbar} \hat{H}_0 t)$ is the time evolution operator associated with $\hat{H}_0$. When $t = 0$, we have

$$|\Psi^{(\text{I})}(0)\rangle = |\Psi^{(\text{S})}(0)\rangle \quad (20)$$

Using the Schrödinger equation for $|\Psi^{(S)}(t)\rangle$, the equation of motion for $|\Psi^{(I)}(t)\rangle$ becomes

$$\boxed{i\hbar \frac{\partial}{\partial t} |\Psi^{(I)}(t)\rangle = \hat{U}_0^\dagger(t) \hat{V} \hat{U}_0(t) |\Psi^{(I)}(t)\rangle = \hat{V}^{(I)}(t) |\Psi^{(I)}(t)\rangle} \quad (21)$$

where the operator in interaction picture is defined as

$$\hat{A}^{(I)}(t) = \hat{U}_0^\dagger(t) \hat{A}^{(S)} \hat{U}_0(t) \quad (22)$$

One can show that the expectation value in the interaction picture is also the same as in Schrödinger picture. The time evolution in the interaction picture can be understood in the following way, **the operator evolves with time via $\hat{H}_0$ and the state evolved with time via $\hat{V}^{(I)}(t)$.**

The formal solution of Eq. (21) can be expressed in terms of the **S-operator**,

$$|\Psi^{(I)}(t)\rangle = \hat{S}(t) |\Psi^{(I)}(0)\rangle \quad (23)$$

Therefore

$$\hat{U}(t) = \hat{U}_0(t) \hat{S}(t) = e^{-\frac{i}{\hbar} \hat{H}_0 t} \hat{S}(t) \quad (24)$$

The expression of $\hat{S}(t)$ can be obtained by iteratively solving Eq. (21) with a perturbative expansion with respect to $\hat{V}^{(I)}(t)$. The final result is:

$$\boxed{\hat{S}(t) = \hat{T} \exp \left\{ -\frac{i}{\hbar} \int_0^t d\tau \hat{V}^{(I)}(\tau) \right\}} \quad (25)$$

where $\hat{T}$ is time ordering operator, if $t_1 > t_2$, $\hat{T}[\hat{V}^{(I)}(t_2) \hat{V}^{(I)}(t_1)] = \hat{V}^{(I)}(t_1) \hat{V}^{(I)}(t_2)$.

3 Quantum Master Equation

3.1 Interaction Picture of the Reduced Density Operator

Recall that in Schrödinger picture, the quantum Liouville equation for $\hat{W}$ reads

$$\frac{\partial}{\partial t} \hat{W}(t) = -\frac{i}{\hbar} [\hat{H}, \hat{W}(t)] \quad (26)$$

and the formal solution is

$$\hat{W}(t) = \hat{U}(t) \hat{W}(0) \hat{U}^\dagger(t) \quad (27)$$

Here we omitted the superscript (S) for quantities in the Schrödinger picture. For the system-reservoir case, $\hat{H}_0$ can be chosen as $\hat{H}_S + \hat{H}_R$ and V as $\hat{H}_{SR}$. Therefore,

$$\hat{U}_0 = \hat{U}_S \hat{U}_R = e^{-\frac{i}{\hbar} \hat{H}_S t} e^{-\frac{i}{\hbar} \hat{H}_R t} \quad (28)$$

$$\hat{S}(t) = \hat{T} \exp \left\{ -\frac{i}{\hbar} \int_0^t d\tau \hat{H}_{SR}^{(I)}(\tau) \right\} \quad (29)$$

Note that $\hat{H}_S$ and $\hat{H}_R$ commute with each other.

By definition, the density operator in the interaction picture is

$$\hat{W}^{(I)}(t) = \hat{U}_0^\dagger(t) \hat{W}(t) \hat{U}_0(t) \quad (30)$$

Substitute this definition into Eq. (26), we have the quantum Liouville equation in interaction picture,

$$\boxed{\frac{\partial}{\partial t} \hat{W}^{(I)}(t) = -\frac{i}{\hbar} [\hat{H}_{\text{SR}}^{(I)}(t), \hat{W}^{(I)}(t)]} \quad (31)$$

Taking partial trace Tr_R on both sides, we get the equation of motion for the RDO in interaction picture:

$$\boxed{\frac{\partial}{\partial t} \hat{\rho}^{(I)}(t) = -\frac{i}{\hbar} \text{Tr}_R \left\{ [\hat{H}_{\text{SR}}^{(I)}(t), \hat{W}^{(I)}(t)] \right\}} \quad (32)$$

where $\hat{\rho}^{(I)}(t) = \text{Tr}_R \hat{W}^{(I)}(t) = \hat{U}_S^\dagger \hat{\rho}(t) \hat{U}_S$.

3.2 Projection Operator

Let $\hat{O}$ be an operator that acts on the total system (both the relevant system and the reservoir), and $\hat{R}$ be an operator that only acts on the reservoir. We define an operator $\hat{P}$ acting on the operator space such that

$$\hat{P}\hat{O} = \hat{R} \text{Tr}_R \hat{O} \quad (33)$$

$\hat{P}$ separate $\hat{O}$ to two parts, one exclusively act on the reservoir ($\hat{R}$) and one exclusively act on the system ($\text{Tr}_R \hat{O}$). We can choose $\hat{R}$ as the equilibrium density operator $\hat{R}_{\text{eq}}$ of the reservoir,

$$\hat{R}_{\text{eq}} = \frac{e^{-\beta \hat{H}_R}}{\text{Tr}_R e^{-\beta \hat{H}_R}} \quad (34)$$

Since $\text{Tr}_R \hat{R}_{\text{eq}} = 1$, we have

$$\hat{P}^2 \hat{O} = \hat{R}_{\text{eq}} \text{Tr}_R \left\{ \hat{R}_{\text{eq}} \text{Tr}_R \hat{O} \right\} = \hat{R}_{\text{eq}} \text{Tr}_R \hat{O} = \hat{P} \hat{O} \quad (35)$$

or $\hat{P}^2 = \hat{P}$. In other words $\hat{P}$ is a projection operator. Its orthogonal complement is defined as

$$\hat{Q} = 1 - \hat{P} \quad (36)$$

If we act $\hat{P}$ and $\hat{Q}$ to the density operator $\hat{W}(t)$, we have

$$\boxed{\hat{P}\hat{W}(t) = \hat{R}_{\text{eq}} \hat{\rho}(t)} \quad (37)$$

$$\boxed{\hat{Q}\hat{W}(t) = \hat{W}(t) - \hat{R}_{\text{eq}} \hat{\rho}(t)} \quad (38)$$

The projector $\hat{P}$ selects the relevant part of the total density operator, which contains the RDO for the relevant system and the equilibrium density operator for the reservoir. The projector $\hat{Q}$ selects less relevant part, such as the fluctuation of the reservoir.

Applying $\hat{P}$ and $\hat{Q}$ to Eq. (31) respectively and using the identity $\hat{W}^{(I)}(t) = \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(t) + \hat{Q}\hat{W}^{(I)}(t)$, we get the following coupled equations for $\hat{\rho}(t)$ and $\hat{Q}\hat{W}^{(I)}(t)$.

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}^{(I)}(t) = -\frac{i}{\hbar}\text{Tr}_R \left\{ [\hat{H}_{\text{SR}}^{(I)}(t), \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(t) + \hat{Q}\hat{W}^{(I)}(t)] \right\}} \quad (39)$$

$$\boxed{\frac{\partial}{\partial t}\hat{Q}\hat{W}^{(I)}(t) = -\frac{i}{\hbar}\hat{Q}[\hat{H}_{\text{SR}}^{(I)}(t), \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(t) + \hat{Q}\hat{W}^{(I)}(t)]} \quad (40)$$

These equation can be solved iteratively using the perturbation expansion with respect to $\hat{H}_{\text{SR}}^{(I)}(t)$. The n th-order equation (n th-order in $\hat{H}_{\text{SR}}^{(I)}(t)$) for $\hat{\rho}^{(I)}(t)$ is

$$\frac{\partial}{\partial t}\hat{\rho}^{(I)}(t) = -\frac{i}{\hbar}\text{Tr}_R \left\{ [\hat{H}_{\text{SR}}^{(I)}(t), \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(t) + \hat{Q}\hat{W}^{(I,n-1)}(t)] \right\} \quad (41)$$

where

$$\hat{Q}\hat{W}^{(I,n-1)}(t) = \hat{Q}\hat{W}^{(I,n-1)}(0) - \frac{i}{\hbar} \int_0^t d\tau \hat{Q}[\hat{H}_{\text{SR}}^{(I)}(\tau), \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(\tau) + \hat{Q}\hat{W}^{(I,n-2)}(\tau)] \quad (42)$$

In the next section we will explore the first and second-order equations for RDO. The first-order approximation leads to the mean-field treatment, and the second-order approximation leads to the quantum master equation.

3.3 First-Order Equation: Mean-Field Approximation

For the most crude approximation, we completely ignore $\hat{Q}\hat{W}^{(I)}(t)$,

$$\hat{Q}\hat{W}^{(I,0)}(t) = 0 \quad (43)$$

Then Eq. (39) becomes first-order in $\hat{H}_{\text{SR}}^{(I)}(t)$,

$$\frac{\partial}{\partial t}\hat{\rho}^{(I)}(t) = -\frac{i}{\hbar}\text{Tr}_R \left\{ [\hat{H}_{\text{SR}}^{(I)}(t), \hat{R}_{\text{eq}}\hat{\rho}^{(I)}(t)] \right\} \quad (44)$$

Since $\hat{\rho}^{(I)}$ only act on system space, it can be taken out of the partial trace. This equation can be simplified as

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}^{(I)}(t) = -\frac{i}{\hbar}[\langle \hat{H}_{\text{SR}}^{(I)}(t) \rangle, \hat{\rho}^{(I)}(t)]} \quad (45)$$

where

$$\langle \hat{H}_{\text{SR}}^{(I)}(t) \rangle = \text{Tr}_R \left\{ \hat{R}_{\text{eq}}\hat{H}_{\text{SR}}^{(I)}(t) \right\} \quad (46)$$

is the thermally averaged **mean-field of system-reservoir coupling**.

Using the relation $\hat{\rho}^{(I)}(t) = \hat{U}_S^\dagger \hat{\rho}(t) \hat{U}_S$, and the facts that $\hat{U}_S$ can be taken out of the partial trace Tr_R , and $\hat{U}_R$ commute with $\hat{R}_{\text{eq}}$, we get the equation of motion of the RDO under mean-field approximation in Schrödinger picture:

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}(t) = -\frac{i}{\hbar}[\hat{H}_S + \langle \hat{H}_{\text{SR}} \rangle, \hat{\rho}(t)]} \quad (47)$$

where $\langle \hat{H}_{\text{SR}} \rangle = \text{Tr}_{\text{R}} \{ \hat{R}_{\text{eq}} \hat{H}_{\text{SR}} \}$. This equation is just the quantum Liouville equation for the system with an additional term in the Hamiltonian accounting for the thermally averaged system-reservoir coupling.

Because the thermally averaged system-reservoir coupling only shifts the system energy scale, **the first-order equation can not describe the dissipation of energy from the system to the reservoir. Such dissipation is caused by the fluctuation of the system-reservoir coupling.**

3.4 Second-Order Equation: Quantum Master Equation

Since $\hat{Q}\hat{W}^{(\text{I},0)}(t) = 0$, the first-order correction to $\hat{Q}\hat{W}^{(\text{I})}(t)$ is

$$\hat{Q}\hat{W}^{(\text{I},1)}(t) = \hat{Q}\hat{W}^{(\text{I},1)}(0) - \frac{i}{\hbar} \int_0^t d\tau \hat{Q} [\hat{H}_{\text{SR}}^{(\text{I})}(\tau), \hat{R}_{\text{eq}} \hat{\rho}^{(\text{I})}(\tau)] \quad (48)$$

We further assume that there is **no initial correlation**, i.e., $\hat{W}^{(\text{I})}(0) = \hat{\rho}^{(\text{I})}(0) \hat{R}_{\text{eq}}$ and $\hat{Q}\hat{W}^{(\text{I})}(0) = 0$. Insert the expression of $\hat{Q}\hat{W}^{(\text{I},1)}(t)$ into Eq. (39), we have the second-order equation of motion for $\hat{\rho}^{(\text{I})}(t)$:

$$\frac{\partial}{\partial t} \hat{\rho}^{(\text{I})}(t) = -\frac{i}{\hbar} \text{Tr}_{\text{R}} \left\{ [\hat{H}_{\text{SR}}^{(\text{I})}(t), \hat{R}_{\text{eq}} \hat{\rho}^{(\text{I})}(t)] \right\} - \frac{1}{\hbar^2} \int_0^t d\tau \text{Tr}_{\text{R}} \left\{ [\hat{H}_{\text{SR}}^{(\text{I})}(t), (1 - \hat{P}) [\hat{H}_{\text{SR}}^{(\text{I})}(\tau), \hat{R}_{\text{eq}} \hat{\rho}^{(\text{I})}(\tau)]] \right\} \quad (49)$$

or

$$\boxed{\frac{\partial}{\partial t} \hat{\rho}^{(\text{I})}(t) = -\frac{i}{\hbar} [\langle \hat{H}_{\text{SR}}^{(\text{I})}(t) \rangle, \hat{\rho}^{(\text{I})}(t)] - \frac{1}{\hbar^2} \int_0^t d\tau \text{Tr}_{\text{R}} \left\{ [\hat{H}_{\text{SR}}^{(\text{I})}(t), (1 - \hat{P}) [\hat{H}_{\text{SR}}^{(\text{I})}(\tau), \hat{R}_{\text{eq}} \hat{\rho}^{(\text{I})}(\tau)]] \right\}} \quad (50)$$

The first term is first-order in $\hat{H}_{\text{SR}}^{(\text{I})}(t)$, which corresponds to the mean-field approximation, and the second term is second-order in $\hat{H}_{\text{SR}}^{(\text{I})}(t)$.

Suppose the system-reservoir coupling can be factorized to system part $\hat{K}_{\mu}(\mathbf{s})$ and reservoir part $\hat{\Phi}_{\mu}(\mathbf{Z})$,

$$\boxed{\hat{H}_{\text{SR}} = \sum_{\mu} \hat{K}_{\mu}(\mathbf{s}) \hat{\Phi}_{\mu}(\mathbf{Z})} \quad (51)$$

This expression is called **multiple factorized form** of the system-reservoir interaction. It is not necessary to require single operators $\hat{K}_{\mu}$ or $\hat{\Phi}_{\mu}$ to be Hermitian. Substitute this expression into Eq. (50), and note the the operators act on different spaces (like $\hat{U}_{\text{S}}$ and $\hat{\Phi}_{\mu}$) commute, the mean-field term becomes:

$$\left(\frac{\partial}{\partial t} \hat{\rho}^{(\text{I})}(t) \right)_{\text{MF}} = -\frac{i}{\hbar} \sum_{\mu} \langle \hat{\Phi}_{\mu} \rangle_{\text{R}} [\hat{K}_{\mu}^{(\text{I})}(t), \hat{\rho}^{(\text{I})}(t)] \quad (52)$$

where $\hat{K}_{\mu}^{(\text{I})}(t) = \hat{U}_{\text{S}}^{\dagger} \hat{K}_{\mu}(t) \hat{U}_{\text{S}}$, $\langle \dots \rangle_{\text{R}} = \text{Tr}_{\text{R}} \{ \hat{R}_{\text{eq}} \dots \}$ indicates that the thermal average is taken for the reservoir.

Next we expand the integrand in the second term. Together with $(1 - \hat{P})$, the double commutator gives rise to 8 terms. For example, one of the terms that does not involve $\hat{P}$ is

$$\begin{aligned}
\text{Tr}_R \left\{ \hat{H}_{\text{SR}}^{(I)}(t) \hat{H}_{\text{SR}}^{(I)}(\tau) \hat{R}_{\text{eq}} \hat{\rho}^{(I)}(\tau) \right\} &= \text{Tr}_R \left\{ \sum_{\mu} \hat{K}_{\mu}^{(I)}(t) \hat{\Phi}_{\mu}^{(I)}(t) \sum_{\nu} \hat{K}_{\nu}^{(I)}(\tau) \hat{\Phi}_{\nu}^{(I)}(\tau) \hat{R}_{\text{eq}} \hat{\rho}^{(I)}(\tau) \right\} \\
&= \sum_{\mu, \nu} \text{Tr}_R \left\{ \hat{\Phi}_{\mu}^{(I)}(t) \hat{\Phi}_{\nu}^{(I)}(\tau) \hat{R}_{\text{eq}} \right\} \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \\
&= \sum_{\mu, \nu} \langle \hat{\Phi}_{\mu}^{(I)}(t) \hat{\Phi}_{\nu}^{(I)}(\tau) \rangle_R \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \\
&= \sum_{\mu, \nu} \langle \hat{\Phi}_{\mu}^{(I)}(t - \tau) \hat{\Phi}_{\nu}^{(I)}(0) \rangle_R \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \quad (53)
\end{aligned}$$

One of the terms that involves $\hat{P}$ is

$$\begin{aligned}
\text{Tr}_R \left\{ \hat{H}_{\text{SR}}^{(I)}(t) \hat{R}_{\text{eq}} \text{Tr}_R \left\{ \hat{H}_{\text{SR}}^{(I)}(\tau) \hat{R}_{\text{eq}} \hat{\rho}^{(I)}(\tau) \right\} \right\} &= \text{Tr}_R \left\{ \hat{H}_{\text{SR}}^{(I)}(t) \hat{R}_{\text{eq}} \right\} \text{Tr}_R \left\{ \hat{H}_{\text{SR}}^{(I)}(\tau) \hat{R}_{\text{eq}} \right\} \hat{\rho}^{(I)}(\tau) \\
&= \sum_{\mu, \nu} \langle \hat{\Phi}_{\mu}^{(I)}(t) \rangle_R \langle \hat{\Phi}_{\nu}^{(I)}(\tau) \rangle_R \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \\
&= \sum_{\mu, \nu} \langle \hat{\Phi}_{\mu} \rangle_R \langle \hat{\Phi}_{\nu} \rangle_R \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \quad (54)
\end{aligned}$$

If we define the **fluctuation operator** as

$$\boxed{\Delta \hat{\Phi}_{\mu}^{(I)}(t) = \hat{\Phi}_{\mu}^{(I)}(t) - \langle \hat{\Phi}_{\mu} \rangle_R} \quad (55)$$

and the **reservoir correlation function** $C_{\mu\nu}(t)$ as

$$\boxed{C_{\mu\nu}(t) = \frac{1}{\hbar^2} \langle \Delta \hat{\Phi}_{\mu}^{(I)}(t) \Delta \hat{\Phi}_{\nu}^{(I)}(0) \rangle_R = \frac{1}{\hbar^2} \left(\langle \hat{\Phi}_{\mu}^{(I)}(t) \hat{\Phi}_{\nu}^{(I)}(0) \rangle_R - \langle \hat{\Phi}_{\mu} \rangle_R \langle \hat{\Phi}_{\nu} \rangle_R \right)} \quad (56)$$

The two examples we discussed can be combined as

$$\hbar^2 \sum_{\mu, \nu} C_{\mu\nu}(t - \tau) \hat{K}_{\mu}^{(I)}(t) \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau) \quad (57)$$

Applying the similar treatment to the other six terms, the integrand becomes

$$\hbar^2 \sum_{\mu, \nu} \left(C_{\mu\nu}(t - \tau) [\hat{K}_{\mu}^{(I)}(t), \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau)] - C_{\nu\mu}(\tau - t) [\hat{K}_{\mu}^{(I)}(t), \hat{\rho}^{(I)}(\tau) \hat{K}_{\nu}^{(I)}(\tau)] \right) \quad (58)$$

and Eq. (50) becomes

$$\boxed{
\begin{aligned}
\frac{\partial}{\partial t} \hat{\rho}^{(I)}(t) &= -\frac{i}{\hbar} \sum_{\mu} \langle \hat{\Phi}_{\mu} \rangle_R [\hat{K}_{\mu}^{(I)}(t), \hat{\rho}^{(I)}(t)] \\
&\quad - \sum_{\mu, \nu} \int_0^t d\tau \left(C_{\mu\nu}(t - \tau) [\hat{K}_{\mu}^{(I)}(t), \hat{K}_{\nu}^{(I)}(\tau) \hat{\rho}^{(I)}(\tau)] \right. \\
&\quad \left. - C_{\nu\mu}(\tau - t) [\hat{K}_{\mu}^{(I)}(t), \hat{\rho}^{(I)}(\tau) \hat{K}_{\nu}^{(I)}(\tau)] \right)
\end{aligned}
} \quad (59)$$

This is the **quantum master equation** (QME) in the interaction picture. QME is a closed equation for $\hat{\rho}^{(1)}(t)$ where all operators act on the system only, while the influence of the reservoir, $\langle\hat{\Phi}_\mu\rangle_{\text{R}}$ and $C_{\mu\nu}(t)$, are just functions. Switching back to Schrödinger picture, the equation becomes

$$\begin{aligned} \frac{\partial}{\partial t}\hat{\rho}(t) = & -\frac{i}{\hbar}[\hat{H}_{\text{S}} + \sum_{\mu}\langle\hat{\Phi}_\mu\rangle_{\text{R}}\hat{K}_\mu, \hat{\rho}(t)] \\ & - \sum_{\mu,\nu}\int_0^t d\tau \left(C_{\mu\nu}(\tau)[\hat{K}_\mu, \hat{U}_{\text{S}}(\tau)\hat{K}_\nu\hat{\rho}(t-\tau)\hat{U}_{\text{S}}^\dagger(\tau)] \right. \\ & \quad \left. - C_{\nu\mu}(-\tau)[\hat{K}_\mu, \hat{U}_{\text{S}}(\tau)\hat{\rho}(t-\tau)\hat{K}_\nu\hat{U}_{\text{S}}^\dagger(\tau)] \right) \end{aligned} \quad (60)$$

The first term involving $\langle\hat{\Phi}_\mu\rangle_{\text{R}}$ leads to reversible dynamics, while the second term involving $C_{\mu\nu}(\tau)$ is responsible for dissipation dynamics. In the second term, the dynamics of $\hat{\rho}(t)$ depends on its dynamic history $\hat{\rho}(t-\tau)$ at all times before t . **The QME is a non-Markovian equation that has memory effects.** Note that the quantum Liouville equation is Markovian. **The non-Markovian behavior is a result of only explicitly describing part of the DOFs of the total system.**